

Projectile Motion with Air Resistance

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1 Physical Model

1.1 Motivation

When we consider a projectile launched from ground level with an initial velocity v_0 at an angle θ above the horizontal, the motion can be described by Newton's second law. If we ask the question of which angle provides the maximum range, in the absence of air resistance, the answer is famously 45° . However this is not the case if we include air resistance.

1.2 The Drag Force

When air resistance is included, the projectile is subject to a drag force $\mathbf{F}_r$ that always acts in the opposite direction of the velocity $\mathbf{v}$.

The form of the drag force depends on the Reynolds number,

$$\text{Re} = \frac{Dv\rho}{\eta}, \quad (1)$$

where D is the projectile's size, ρ the density of air, and η its viscosity.

Re is the ratio of inertial to viscous forces, inside of a moving fluid. If we have a low Re, viscous forces dominate and the drag force is linear,

$$F_r \propto v \quad (\text{linear, or Stokes, drag}), \quad (2)$$

which is a regime typical of small objects moving slowly through a viscous fluid, such as a droplet.

On the other hand, if we have a high Re, inertial forces dominate and the drag force is quadratic,

$$F_r \propto v^2 \quad (\text{quadratic, or Newtonian, drag}), \quad (3)$$

which is a regime typical of bigger objects moving through air, such as a thrown ball or a fired projectile.

2 System of Differential Equations

3 Numerical Integration

4 Stopping Condition

5 How this turns into Code